

The Scales of Harmonies

An overview of 33 scales in 7 families, by **Oliver Prehn** (NewJazz) — colour-coded study sheet

The 33 scales presented in the NewJazz lesson are organised in 7 families. Each family has one interval pattern; a scale is that pattern read from a given degree — which is why the rows of a family line up as a staircase. The 33 scales follow from a few simple rules:

- Only half, whole and whole-and-a-half steps
- Half steps may not be neighbours
- Whole and whole-and-a-half steps may not be neighbours
- Whole-and-a-half steps may not be neighbours

Following them limits the total to 33 scales, all well suited to building clear, evident harmonies.

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Reading the colours



• **Blue** and **red** are the two whole-tone sets of the keyboard. A **whole step (1)** stays in the same colour; a **half step (½)** or a **step and a half (1½)** crosses to the other. Each column of the grid is one semitone.

FAMILY	POPULAR NAME	INTERVAL STEPS	*SYSTEMATIC NAME	CHORDS
1	Major		Ionian	Ma7
	Dorian		Dorian	mi7
	Phrygian		Phrygian	mi7
	Lydian		Lydian	Ma7
	Mixolydian		Mixolydian	7
	Natural minor		Aeolian	mi7
	Locrian		Locrian	mi7b5
2	Altered / Super Locrian		Ionian #1	7alt / mi7b5
	Ascending mel. minor		Dorian #7	miMa7
	Dorian b2		Phrygian #6	mi7
	Lydian Augmented		Lydian #5	Ma7#5
	Lydian dominant		Mixolydian #4	7
	Aeolian dominant		Aeolian #3	7
	Half diminished		Locrian #2	mi7b5
3	Major #5 / Major Aug.		Ionian #5	Ma7#5
	Dorian #4		Dorian #4	mi7
	Phrygian dominant		Phrygian #3	7
	Lydian #2		Lydian #2	Ma7
	Altered dominant bb7		Mixolydian #1	dim7
	Harmonic minor		Aeolian #7	miMa7
	Locrian #6		Locrian #6	mi7b5
4	Harmonic Major		Ionian b6	Ma7
	Dorian b5		Dorian b5	mi7b5
	Phrygian b4		Phrygian b4	mi7 / 7
	Lydian b3		Lydian b3	miMa7
	Mixolydian b2		Mixolydian b2	7
	Lydian augmented #2		Aeolian b1	Ma7#5 / dim7
	Locrian bb7		Locrian b7	dim7
5	Diminished		Diminished	dim7
	Dominant diminished		Inverted diminished	7
6	Whole tone		Whole tone	7 #5/b5
7	Augmented		Augmented	Ma7
	Inverted Augmented		Inverted Augmented	6#5

*The systematic names do not follow the jazz-school habit of naming every scale after the Major scale. "Aeolian #7" is an Aeolian scale with a raised 7th

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